Assignment # 5

STAT 679: Statistical Methods for Spatial Lattice Data–Spring 2026, UW-Madison

Cumulative Homework #5

For your final homework assignment, you will undertake a paper replication of a spatial analysis.[1] Please carefully read the Background, Methods, and Results section of a spatial analysis below. Your assignment will be to replicate the results and perform additional analyses. The data will be provided to you on Canvas (`wiag_sf`). The variable definitions are below:

- `GEOID`: partial geographic identifier
- `NAME`: county name
- `cr_2012`: corn yield (bushels/acre) in 2012
- `cr_2013`: corn yield (bushels/acre) in 2013
- `pr_2012`: precipitation in 2012 (in inches)
- `sy_2012`: soybean yield (bushels/acre) in 2012
- `sy_2013`: soybean yield (bushels/acre) in 2013

Background

As the climate changes, farmers throughout the United States are under increasing pressure to adapt. Increasing rainfall and moisture make conditions challenging to harvest both corn and soybean (Kirwan, n.d.a, n.d.b). Compared to corn, however, the soybean yield potential of late planting decreases slower than that of corn (Lauer 2002). Therefore, it is of interest to farmers to better understand the relationship between soybean yield in a current year and soybean and corn yield in the previous years.

Soybean and corn respond similarly to environmental factors and generally seem to follow each other across time. In 2013, soybean yield across Wisconsin decreased, which was preceded by a dip in corn yield the year before. More recently, between 2017 and 2018 soybean yield appears to be increasing where corn yield appears to be decreasing. If the previous relationship in 2012 and 2013 can be better characterized, farmers hope that they will be better able to plan and understand how soybean and corn trend together.

The goal of this analysis is to quantify the relationship between soybean yield in 2013 and corn yield in 2012 and soybean yield in 2012. We adjust for spatial dependence at the county level.

Methods

Soybean and corn yield data were obtained from the USDA (United States Department of Agriculture) National Agricultural Statistics Service at the county level for Wisconsin. Yield for both

[1] This example was developed for educational purposes only.

crops is measured in bushels per acre. Across 2012 and 2013, some county-level yields were not published by the USDA in order to avoid disclosing the yield of individual agricultural operations. For these counties, yield values were imputed using predictive mean matching. A table of summary statistics can be found in Table 1.

Table 1: Summary Statistics for Wisconsin Corn/Soybean Data

		Min.	Q1	Median	Mean	Q3	Max
Wisconsin	Corn 2012	60.0	92.50	110.80	108.58	121.50	167.1
Wisconsin	Corn 2013	81.9	117.92	128.80	129.71	141.35	179.3
Wisconsin	Soybean 2012	28.30	39.08	43.82	43.64	49.30	56.50
Wisconsin	Soybean 2013	18.00	26.48	30.48	31.24	36.44	51.60

In order to evaluate complete spatial randomness, we use Geary's C to test for spatial autocorrelation in corn yield across Wisconsin in 2012 and soybean yield across Wisconsin in 2013. We use binary spatial weights. For soybean yield in 2013, Geary's C statistic was 0.475 with standard deviate 6.282 and p-value < 0.001, suggesting strong positive spatial correlation. For corn yield in 2012, Geary's C statistic was 1.045 with standard deviate -0.541 and p-value 0.705, suggesting no evidence of positive spatial correlation in corn yield in 2012.

Results

After performing standard linear regression (OLS), we identified evidence of residual spatial autocorrelation using Moran's I test on the residuals by randomization. The Moran's I statistic for the OLS residuals was 0.314, with standard deviate 4.683 and p-value < 0.001. We therefore use a first-order simultaneous autoregressive (SAR) error model, which adequately adjusts for the spatial dependence. We use binary spatial weights. A comparison between the OLS and SAR models can be found in Table 2.

Table 2: Model Comparisons

	OLS	SAR
Corn 2012	0.100**	0.044
	(0.031)	(0.024)
Soybean 2012	0.584***	0.440***
	(0.111)	(0.083)
Constant	-5.128	9.659*
	(5.988)	(5.016)
Observations	72	72
R^2	0.353	–
Adjusted R^2	0.334	–
Lambda	–	0.1329 (LR Test Value: 20.691***)
Log-likelihood	–	-221.528
ML Residual Variance (σ^2)	–	24.139
AIC	471.74	453.06
Residual Standard Error	6.189 ($df = 69$)	–
F-statistic	18.81*** ($df = 2; 69$)	–
Moran's I Statistic: Residuals	0.314***	-0.024

**p<0.05; ***p<0.01

Conclusions

Based on our results, the standard OLS model suggested that both corn yield in 2012 and soybean yield in 2012 were associated with soybean yield in 2013 in Wisconsin. In the OLS model, corn yield in 2012 had a positive and statistically significant association with soybean yield in 2013, and soybean yield in 2012 also had a positive and statistically significant association with soybean yield in 2013.

However, the OLS residuals showed strong evidence of positive spatial autocorrelation, indicating that the independence assumption was violated. After adjusting for residual spatial autocorrelation using the SAR error model, the estimated effect of corn yield in 2012 was reduced substantially and was no longer significant at the 5% level, although it remained marginally significant at the 10% level. Soybean yield in 2012 remained strongly associated with soybean yield in 2013. The SAR model also removed the residual spatial autocorrelation, suggesting that the spatial error structure better accounts for dependence among neighboring counties. These results suggest that ignoring spatial autocorrelation can affect inference about the relationship between previous-year corn yield and current-year soybean yield.

Assignment

1. **(10 pts)** Perform an exploratory spatial data analysis (ESDA) on the corn and soybean data and how yield differs by county. Include a different visualization you use as part of your ESDA and interpret what you observe.

2. **(10 pts)** The authors use Geary's c to test for spatial dependence in corn yield in 2012 and

in soybean yield in 2013. Use Moran's I instead to test for spatial dependence in corn yield in 2012 and soybean yield in 2013 and interpret your findings. Do you come to the same conclusions?

3. **(20 pts)** The authors report estimates using both standard linear regression analysis results (OLS) and the first order simultaneous autoregressive error model (SAR).

   (a) Replicate their analysis and findings for the linear regression. Test the model residuals for spatial correlation. Show your outputs and include your code.

   (b) Replicate their analysis instead using the CAR error model. Do your results differ? If so, how? Do you prefer the CAR model, SAR model, or neither? Support your answer. Show your model output and include your code.

4. **(20 pts)** The authors discuss the importance of precipitation yet do not include it in their dataset. In your spatialpolygonsdataframe (`wiag_spdf`), we have included the (synthetic) mean precipitation by county in Wisconsin for 2012.

   (a) Adjust for precipitation in the SAR model by including mean precipitation for each county in 2012 as a predictor (variable name: `pr_2012`). Show your summary results and include your code. Test for residual spatial autocorrelation. Interpret the coefficients from the SAR model.

   (b) Adjust for precipitation in the CAR model by including mean precipitation for each county in 2012 as a predictor (variable name: `pr_2012`). Show your summary results and include your code. Test for residual spatial autocorrelation. Interpret the coefficients from the CAR model.

5. **(5 pts)** Based on your findings, what additional information would you want to explore before concluding your findings? Do you agree with the authors in their final conclusions?

References

Kirwan, Hope. n.d.a. *Wisconsin Corn Farmers Facing More Market Uncertainty After USDA Planting Report*. Wisconsin Public Radio. `https://www.wpr.org/wisconsin-corn-farmers-facing-more-market-uncertainty-after-usda-planting-report`.

n.d.b. *Wisconsin Farmers 'Getting Hit Big Time' as Weather Delays Corn Planting and Decisions about Crop Loom*. Wisconsin Public Radio. `https://www.wpr.org/wisconsin-farmers-getting-hit-big-time-weather-delays-corn-planting-and-decisions-about-crop-loom`.

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